

Deep Learning Techniques for COVID-19 Detection from Chest X-rays

Chest X-rays have been investigated for computer-aided analysis of respiratory disease, including COVID-19.

CNN architectures can learn visual patterns from chest X-ray images and classify images into predefined categories.

Transfer learning with pretrained image models can be useful when medical datasets are relatively small.

Evaluation measures include sensitivity, specificity, precision, recall, F1 score, accuracy, and ROC-AUC.

Challenges include heterogeneous datasets, equipment differences, population differences, class imbalance, and model interpretability.

AI analysis should be treated as decision-support research unless appropriately validated and authorized.